

CS105 (DIC on Discrete Structures)

Problem set 8

- Attempt *all* questions.
 - Apart from things proved in lecture, you cannot assume anything as “obvious”. Either quote previously proved results or provide clear justification for each statement.
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Part 1

1. The drama club has m members and the dance club has n members. For the upcoming paf, a committee of k people needs to be formed with at least one member from each club. If the clubs have exactly r members in common, what is the number of ways the committee may be chosen? Justify your answer.
2. Let $B(n)$ be the no. of bit strings of length n that contain the string 01.
 - (a) Write a recurrence relation for $B(n)$.
 - (b) Determine the initial conditions.
 - (c) How many strings are there of length 7 are there that contain 01?
 - (d) Solve the recurrence to obtain an expression for $B(n)$ in terms of n .
3. Consider the standard deck of 52 playing cards. A balanced hand is a subset of 13 cards containing four cards of one suit and three cards of each of the remaining three suits. Find N , the number of balanced hands. Find the number of ways of dealing the cards to four (distinguishable) players so that each player gets a balanced hand. Is this number equal to $N(N-1)(N-2)(N-3)$?
4. Find the number of integral solutions for $x_1 + x_2 + x_3 + x_4 + x_5 = 21$ with each $x_i \geq 2$. Next, generalize this to find the number of solutions for $\sum_{i=1}^k x_i = n$ with each $x_i \geq t$ and express it in a closed form involving n , t , and k .

Part 2

5. Two hundred students participated in a math contest. They had six problems to solve. Each problem was correctly solved by at least 120 participants. Prove, using double counting, that there must be two participants such that every problem was solved by at least one of these students.
6. Consider a grid from $(0,0)$ to (n,n) . Starting from the point $(0,0)$, we wish to take units steps ONLY in the direction of the positive X and Y axes (i.e., right and up), and reach the point (n,n) . Find a recurrence relation for the number of ways of doing so, if we are not allowed to go above the line joining $(0,0)$ and (n,n) , i.e, the diagonal.
7. Consider the number of strings x_n of length n over the set $\{0,1,2\}$ such that any two consecutive numbers in the string differ by at most 1. Thus 01221 is such a string of length 5, but 01200 is not.
 - (a) Write a recurrence relation for x_n .
 - (b) What are x_1 , x_2 and x_7 ?
 - (c) Solve the recurrence to get a closed form solution for x_n in terms of n .